Practice Quiz: Lecture 3 – Part 2

Topics: Analog Filters, Impairments (Attenuation, Noise, Distortion), Decibels & dBm, Nyquist Noiseless Bit Rate, Shannon Channel Capacity, Propagation Delay & Throughput

Student Name: _____ **Student ID:** _____ **Score:** _____ / 20

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1. Match each filter type with its corresponding operational description:

Description	Filter Type Options
1. Allows only signals between cutoff frequencies f_1 and f_2 to pass.	[Low-pass / High-pass / Band-pass]
2. Used primarily to eliminate high-frequency noise from a baseband signal.	[Low-pass / High-pass / Band-pass]
3. Blocks low-frequency DC bias while passing higher transmission frequencies.	[Low-pass / High-pass / Band-pass]

2. Identify the appropriate transmission impairment terminology for each statement:

Statement	Terminology
1. Signal power progressively decreases as it propagates over physical distance.	_____
2. Different frequency components travel at different speeds, changing signal shape.	_____
3. Unwanted external electromagnetic signals corrupt the transmitted data.	_____

3. A signal has an initial power of $P_1 = 10\ \text{mW}$. After traversing a transmission medium, its power drops to $P_2 = 0.5\ \text{mW}$. Calculate the power change in decibels (dB).
(Assume** $\log_{10}(2) \approx 0.3$ **and** $\log_{10}(5) \approx 0.7$**)

Answer: _____

4. A signal enters an amplifier at $P_1 = 8\ \mathrm{mW}$ and exits at $P_2 = 2\ \mathrm{mW}$. Express the attenuation in decibels (dB).
(Assume** $\log_{10}(2) \approx 0.3$**)

Answer: _____

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5. Convert the following power values between Watts (W), milliwatts (mW), and dBm:

a) $1\ \mathrm{W} =$ _____ $\mathrm{mW} =$ _____ dBm

b) $-20\ \mathrm{dBm} =$ _____ mW

c) $0\ \mathrm{dBm} =$ _____ mW

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6. A radio transmitter has an output power of $5\ \mathrm{W}$. Using the approximation $\log_{10}(5) \approx 0.7$, convert this output power into dBm.

Answer: _____

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7. A signal with an initial power of $16\ \mathrm{mW}$ passes through a transmission channel exhibiting a net loss of $-9\ \mathrm{dB}$. Given $\log_{10}(2) \approx 0.3$, calculate the output signal power in milliwatts (mW).

Answer: _____

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8. A composite transmission link consists of three cascaded stages: a cable section with $-14\ \mathrm{dB}$ loss, an intermediate amplifier providing $+20\ \mathrm{dB}$ gain, and a second cable section with $-12\ \mathrm{dB}$ loss. Calculate the overall decibel change of the link.

Answer: _____

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9. A communication channel has a bandwidth of $B = 5000\ \text{Hz}$ and uses $L = 8$ distinct signal levels. Assume that channel noise can be completely neglected.

a) Which theoretical theorem must be used to calculate the maximum bit rate? _____

b) What is the maximum data rate in kbps? _____

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10. A noisy channel has a bandwidth of $B = 10\ \text{kHz}$ and a linear signal-to-noise ratio of $\text{SNR} = 15$.

a) Which theoretical theorem defines the absolute channel capacity? _____

b) What is the maximum channel capacity in kbps? _____

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11. A telephone channel has an allocated bandwidth of $B = 4\ \text{kHz}$ and an $\text{SNR} = 63$. Calculate the Shannon channel capacity in kbps.

Answer: _____

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12. An optical channel has a bandwidth of $B = 3\ \text{kHz}$ and operates with a measured signal-to-noise ratio in decibels of $\text{SNR}_{\text{dB}} = 30\ \text{dB}$. What is its theoretical Shannon capacity in kbps? *(Hint:* $2^{10} = 1024 \approx 1000$*)*

Answer: _____

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13. A network transmits $6000\ \text{frames}$ per minute. Each transmitted frame carries exactly $10,000\ \text{bits}$. What is the effective network throughput in Mbps?

Answer: _____

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14. A telemetry link transmits $12,000\ \text{frames}$ per minute, where each frame consists of $2,500\ \text{bytes}$ (1 byte = 8 bits). Calculate the throughput in Mbps.

Answer: _____

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15. A signal propagates a distance of $d = 2400\ \text{km}$ through a cable at a propagation velocity of $v = 2.4 \times 10^8\ \text{m/s}$. Calculate the propagation time (T_p) in milliseconds (ms).

Answer: _____

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16. An undersea fiber-optic cable spans $d = 3600\ \text{km}$. Light propagates through the glass core at $v = 2.0 \times 10^8\ \text{m/s}$. What is the total one-way propagation delay in milliseconds (ms)?

Answer: _____

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17. Which three fundamental physical factors determine the achievable maximum data rate of a physical communication channel?

( ) A. Cable color, socket pinning, and operating system kernel version.

( ) B. Channel bandwidth, number of signaling levels, and channel quality (SNR).

( ) C. MAC header format, IP subnet mask, and DNS lookup delay.

( ) D. Packet frame size, application payload type, and window buffer count.

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18. Contrast how attenuation is restored in transmission: What device is used to regenerate digital signals versus restoring analog signals?

( ) A. Analog signals use repeaters (regenerators); digital signals use passive attenuators.

( ) B. Analog signals use amplifiers (which amplify both signal and noise); digital signals use repeaters (which reconstruct clean binary pulses).

( ) C. Both digital and analog links use low-pass filters exclusively.

( ) D. Digital signals require oscilloscopes to restore line attenuation.

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19. A network link has a bandwidth of $B = 3000\ \text{Hz}$ and an $\text{SNR} = 31$. According to Shannon's theorem, its capacity is $15\ \text{kbps}$. To achieve this theoretical rate on an equivalent noiseless channel using Nyquist's theorem, how many discrete signal levels (L) must be supported?

Answer: _____

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20. A packet transmission exhibits the following measured latencies: Transmission delay $T_{\text{trans}} = 4\ \text{ms}$, Propagation delay $T_p = 12\ \text{ms}$, Queuing delay $T_{\text{queue}} = 3\ \text{ms}$, and Processing delay $T_{\text{proc}} = 1\ \text{ms}$. What is the total latency experienced by the packet?

Answer: _____

Answer Key & Detailed Explanations (Lecture 3 – Part 2)

Question 1: Matching

1. Allows frequencies between f_1 and f_2 to pass $\rightarrow$ **Band-pass filter**
2. Removes high-frequency noise from baseband $\rightarrow$ **Low-pass filter**
3. Blocks low-frequency DC bias while passing higher frequencies $\rightarrow$ **High-pass filter**

Question 2: Terminology

1. Loss of signal energy over distance $\rightarrow$ **Attenuation**
2. Differing velocity/delay across frequency components $\rightarrow$ **Distortion (Delay Distortion)**
3. Extraneous external interference $\rightarrow$ **Noise**

Question 3: -13 dB

$$\text{Gain/Loss (dB)} = 10\log_{10}\left(\frac{P_2}{P_1}\right) = 10\log_{10}\left(\frac{0.5}{10}\right) = 10\log_{10}\left(\frac{5}{100}\right)$$

$$= 10\left[\log_{10}(5) - \log_{10}(100)\right] = 10(0.7 - 2) = 10(-1.3) = -13 \text{ dB}$$

Question 4: -6 dB (or an attenuation of 6 dB)

$$\text{Loss (dB)} = 10\log_{10}\left(\frac{P_2}{P_1}\right) = 10\log_{10}\left(\frac{2}{8}\right) = 10\log_{10}\left(\frac{1}{4}\right) = 10\log_{10}(2^{-2})$$

$$= 10 \times (-2) \times \log_{10}(2) = -20 \times 0.3 = -6 \text{ dB}$$

Question 5: Power Conversions

a) $1\text{ W} = 1000\text{ mW} = 10\log_{10}(1000) = \mathbf{30\text{ dBm}}$

b) $-20\text{ dBm} \implies P_{\text{mW}} = 10^{-20/10} = 10^{-2} = \mathbf{0.01\text{ mW}}$

c) $0\text{ dBm} \implies P_{\text{mW}} = 10^{0/10} = 10^{0} = \mathbf{1\text{ mW}}$

Question 6: 37 dBm

First express power in milliwatts: $5\text{ W} = 5000\text{ mW}$.

$$P_{\text{dBm}} = 10\log_{10}(5000) = 10\left[\log_{10}(5) + \log_{10}(1000)\right] = 10(0.7 + 3) = 10(3.7) = 37 \text{ dBm}$$

Question 7: 2 mW

$$-9\text{ dB} = 10\log_{10}\left(\frac{P_2}{16}\right) \implies \log_{10}\left(\frac{P_2}{16}\right) = -0.9 = -3 \times 0.3 = \log_{10}(2^{-3}) = \log_{10}\left(\frac{1}{8}\right)$$

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$$\frac{P_2}{16} = \frac{1}{8} \implies P_2 = \frac{16}{8} = 2 \text{ mW}$$

Question 8: -6 dB

Decibels can be algebraically summed across cascaded stages:

$$\text{Net dB} = (-14 \text{ dB}) + (+20 \text{ dB}) + (-12 \text{ dB}) = -6 \text{ dB}$$

Question 9: a) Nyquist Theorem, b) 30 kbps

Nyquist's formula applies to noiseless channels:

$$R_{\max} = 2B \log_2(L) = 2 \times 5000 \times \log_2(8) = 10000 \times 3 = 30000 \text{ bps} = 30 \text{ kbps}$$

Question 10: a) Shannon Capacity Theorem, b) 40 kbps

Shannon's formula determines the upper bound for channels with noise:

$$C = B \log_2(1 + \text{SNR}) = 10000 \times \log_2(1 + 15) = 10000 \times \log_2(16) = 10000 \times 4 = 40000 \text{ bps} = 40 \text{ kbps}$$

Question 11: 24 kbps

$$C = B \log_2(1 + \text{SNR}) = 4000 \times \log_2(1 + 63) = 4000 \times \log_2(64) = 4000 \times 6 = 24000 \text{ bps} = 24 \text{ kbps}$$

Question 12: $\approx$ 30 kbps

Convert SNR from dB to linear: $\text{SNR} = 10^{30/10} = 10^3 = 1000$.

$$C = B \log_2(1 + 1000) \approx 3000 \times \log_2(1024) = 3000 \times 10 = 30000 \text{ bps} = 30 \text{ kbps}$$

Question 13: 1 Mbps

$$\text{Throughput} = \frac{\text{Total bits}}{\text{Time (seconds)}} = \frac{6000 \text{ frames} \times 10000 \text{ bits}}{60 \text{ s}} = \frac{60,000,000 \text{ bits}}{60 \text{ s}} = 1,000,000 \text{ bps} = 1 \text{ Mbps}$$

Question 14: 4 Mbps

Convert bytes to bits: $2500 \text{ bytes} \times 8 = 20,000 \text{ bits/frame}$.

$$\text{Throughput} = \frac{12000 \times 20000 \text{ bits}}{60 \text{ s}} = \frac{240,000,000 \text{ bits}}{60 \text{ s}} = 4,000,000 \text{ bps} = 4 \text{ Mbps}$$

Question 15: 10 ms

$$T_p = \frac{\text{Distance}}{\text{Velocity}} = \frac{2400 \times 10^3 \text{ m}}{2.4 \times 10^8 \text{ m/s}} = \frac{2.4 \times 10^6}{2.4 \times 10^8} = 0.01 \text{ s} = 10 \text{ ms}$$

Question 16: 18 ms

$$T_p = \frac{d}{v} = \frac{3600 \times 10^3 \text{ m}}{2.0 \times 10^8 \text{ m/s}} = \frac{3.6 \times 10^6}{2.0 \times 10^8} = 0.018 \text{ s} = 18 \text{ ms}$$

Question 17: B

Achievable channel capacity is fundamentally determined by the channel's bandwidth, the number of discrete signaling levels supported by the transceivers, and the quality of the channel (noise level / SNR).

Question 18: B

Amplifiers boost analog signal amplitude along with any accumulated noise. In contrast, digital repeaters decode the pulses and regenerate brand-new, noise-free clean square waveforms.

Question 19: $L = 6$ levels **(or** $L \geq 8$ **in power-of-2 systems)**

Equating Shannon capacity to Nyquist rate:

$$C = 2B \log_2(L) \implies 15000 = 2(3000) \log_2(L) \implies 6000 \log_2(L) = 15000$$

$$\log_2(L) = \frac{15000}{6000} = 2.5 \implies L = 2^{2.5} = 2^2 \times 2^{0.5} \approx 4 \times 1.414 \approx 5.66$$

Thus, the system theoretically requires at least 6 levels ($L = 8$ in practice).

Question 20: 20 ms

$$\text{Total Latency} = T_{\text{proc}} + T_{\text{queue}} + T_{\text{trans}} + T_p = 1 \text{ ms} + 3 \text{ ms} + 4 \text{ ms} + 12 \text{ ms} = 20 \text{ ms}$$
